Lab 1: Coulomb's Law

In the late 1700's, Charles Coulomb discovered that the force that two spheres of electric charge exert on each other has the same form as Newton's description of *universal gravitation*… that is, the force is proportional to the product of the two charges and inversely proportional to the square of the distance between them:

$$F = \frac{GMm}{r^2}$$ *Newton's Law of Universal Gravitation*

$$F = \frac{kq_1q_2}{r^2}$$ *Coulomb's Law (for two spheres of charge)*

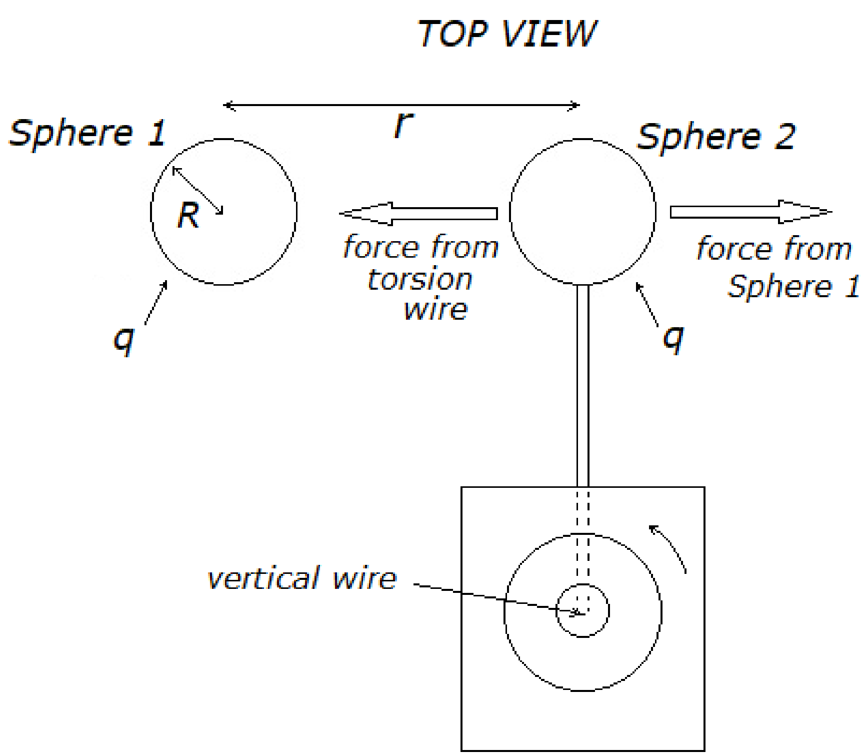
where q_1 and q_2 are the magnitude of the two charges and k is Coulomb's constant, and r is the distance between the centers of the two spheres.

In this experiment, we will test Coulomb's Law two ways:

Trial 1: We will hold r constant while varying the value of q

Trial 2: We will hold q constant while varying the value of r

Our apparatus includes one fixed sphere (labeled "Sphere 1") and one sphere (labeled "Sphere 2") suspended from a thin vertical wire that will "twist" when a force is applied to the sphere. We can twist the wire the opposite direction to return the sphere to its equilibrium position, so that the force applied by the wire will balance the force applied by the fixed sphere:



In this way, the force acting on Sphere 2 will be measured indirectly, **as an angle through which the wire is twisted.** We can reasonably assume that the force applied will be proportional to the angle, so that:

$$F = \alpha\,\theta$$

Where θ is the angle through which the wire is twisted and α is the constant of proportionality.

The charge on both spheres will be identical for each measurement, and this charge will also be determined indirectly. The spheres will be "charged" using a probe attached to a high-voltage power supply, so that the electric potential of each sphere will be set to the value of the power supply. In this way, we can relate the potential of the sphere, V, to the charge on the sphere and the radius of the sphere:

$$V = \frac{kq}{R}$$

In this expression, q is the charge on the sphere, R is the radius of the sphere and k is Coulomb's constant.

We can now use these two expressions to modify Coulomb's Law so we can create a version that will be useful for the data we will measure:

$$F = \frac{kqq}{r^2}$$

$$\alpha\,\theta = \frac{k(RV/k)^2}{r^2}$$

$$\theta = \frac{R^2\,V^2}{k\,\alpha\,r^2}$$

Note that there are two constants in this expression, k and α, and four elements that we will measure:

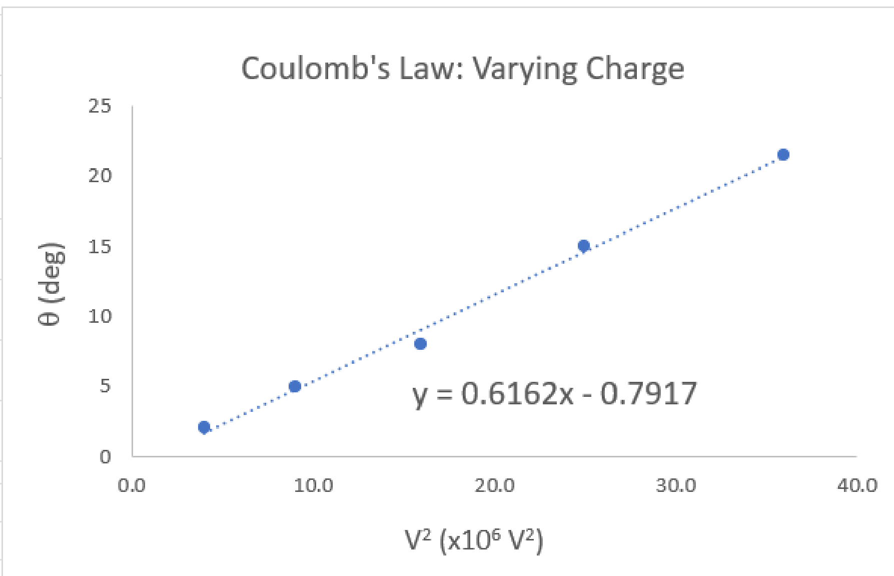
R: the radius of a sphere
V: the potential of the power supply, used to "charge" the spheres
r: the distance between the centers of the spheres
θ: the angle the wire is twisted to keep the forces on Sphere 2 in equilibrium

Part 1: hold *r* constant, vary *V* and *θ*

For Part 1, we will hold ***r*** constant at 6.00 cm, and then measure the angle for five values of ***V***. Refer to the video for the measured data and create a data table and graph for Part 1 that looks like this:

NOTE: the data for θ shown here is NOT the actual data

Part 1: Vary Charge	r = 6.00 cm	
V (kV)	V^2 ($\times 10^6$ V^2)	θ (deg)
2.0	4.0	2.0
3.0	9.0	5.0
4.0	16.0	8.0
5.0	25.0	15.0
6.0	36.0	21.5



We can modify the last equation derived above:

$$\theta = \frac{R^2\, V^2}{k\, \alpha\, r^2} \qquad or \qquad \theta = \beta\ V^2 \qquad where \qquad \beta = \frac{R^2}{k\, \alpha\, r^2}$$

Since your graph is of ***q*** vs ***V²*** the slope of the graph is represented by *β*. By using the value of the slope of the graph, along with values of q r and R, we can calculate the experimental value of Coulomb's constant:

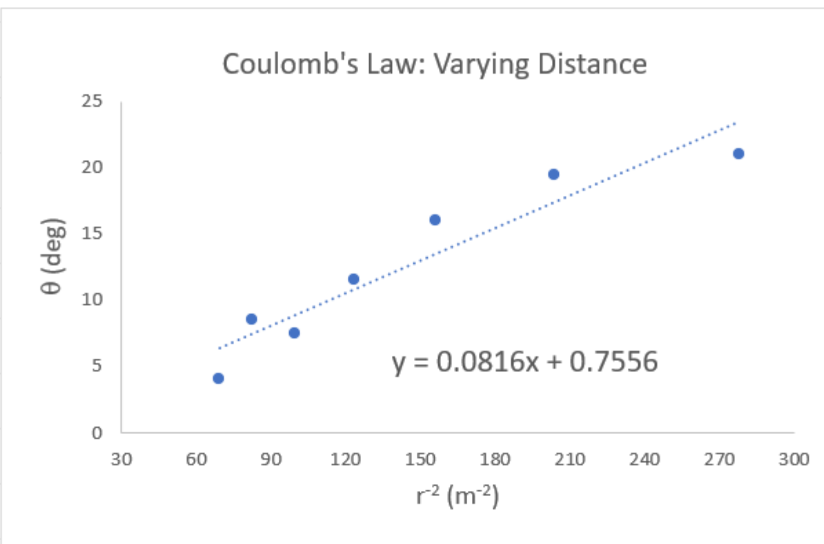
$$k = \frac{R^2}{\alpha\, \beta\, r^2}$$

(Note: we will determine the value of α in Part 3)

Part 2: hold *V* constant, vary *r* and θ

For Part 2, we will hold *V* constant at 6000 Volts, and then measure the angle for seven values of *r*. Refer to the video for the measured data and create a data table and graph for Part 2 that looks like this:

Part 2: Vary Distance	V = 6.0 kV	
r (cm)	r^{-2} (m^{-2})	θ (deg)
12.0	69	4.0
11.0	83	8.5
10.0	100	7.5
9.0	123	11.5
8.0	156	16.0
7.0	204	19.5
6.0	278	21.0



Once again, we can modify our equation derived above:

$$\theta = \frac{R^2\, V^2}{k\, \alpha\, r^2} \qquad or \qquad \theta = \gamma\ r^{-2} \qquad where \qquad \gamma = \frac{R^2\, V^2}{k\, \alpha}$$

Since your graph is of ***q*** vs ***r*** $^{-2}$ the slope of the graph is represented by γ. By using the value of the slope of the graph, along with values of q r and R, we can calculate the experimental value of Coulomb's constant:

$$k = \frac{R^2\, V^2}{\alpha\, \gamma}$$

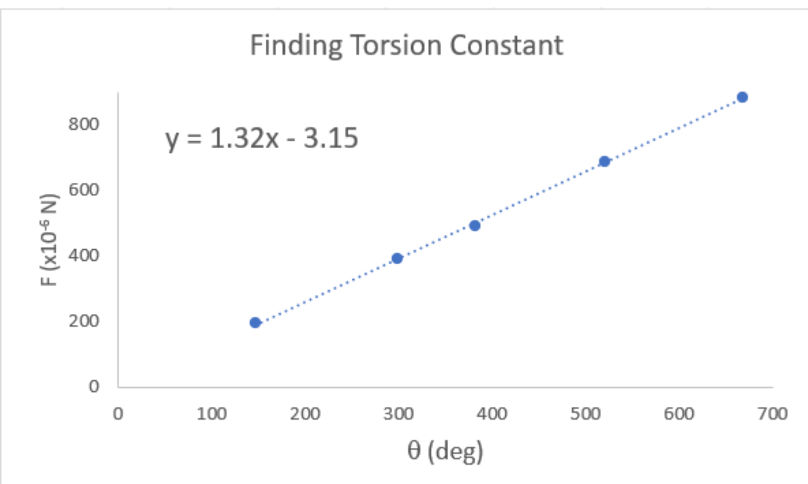
(Note: we will determine the value of α in Part 3)

Part 3: determine the force constant for the wire, using known forces

To determine the force constant – i.e. the constant of proportionality α – between the angle the wire is twisted and the force it applies to Sphere 2, we will turn the apparatus on its side and place a small metal mass on top of the sphere. The weight of the mass will push down on the sphere, and the wire will be twisted to apply a force to the sphere in the opposite direction. We will collect data – mass and angle – for five values. Create a data table and graph like this:

Part 3: Finding Torsion Constant

m ($\times 10^{-6}$ kg)	θ (deg)	F ($\times 10^{-6}$ N)
20	147	196
40	299	392
50	382	490
70	521	686
90	668	882



When we apply our equation at the top of Page 2, $F = \alpha\,\theta$ to the results of this graph, we can see that:

the slope of this graph is the value of the constant α

Finally… you can now use this value of α, along with the values of the other constants, to calculate *k*, Coulomb's constant, for Part 1 and Part 2.